Seat No._____ Student number_____ Name_____ Major_____ Grade_____

装 订 线

Stony Brook Institute at Anhui University

The First Semester of 2022-2023

Probability Theory and Mathematical Statistics

Middle exam

(Closed book Time 120 minutes)

Question number	I	II	III	IV	Total score
Score					
Marker					

I. Completions (5 items, 2 points each)

Score	

1. Event A is a subset of event B, and $P(A) = 1/6$, $P(B) = 1/3$, then $P(\bar{A} \cap B)$ is_____.

2. Suppose that the pdf of r.v. X is $f(x) = \begin{cases} 3x^2 & 0 \le x < 1 \\ 0 & \text{others} \end{cases}$, then the probability of X falls in interval (a, b) is_____, where $0 < a < b < 1$.

3. Suppose that r.v. $X \sim U(0,5)$, then the probability that equation $t^2 + X - 2 = 0$ has at least a real root is_____.

4. Suppose that the number of defective products in a bag obeys a Poisson distribution with $\lambda = 3$, then the probability that no defective product in the bag is_____.

5. Suppose that r.v. X obeys a binomial distribution with parameters of $2n + 1$ and 0.5, n is a non-negative integer. Then the probability that X is larger than n is_____.

II. Choices (5 items, 2 points each)

Score	

6. Given that $P(A) = 0.4$, $P(B) = 0.6$, $P(A \cup B) = 0.9$, then $P(A \cap \bar{B})$ is ( ).

   A. 0.3 B. 0.4
   C. 0.6 D. 0.7

7. Suppose that A and B are two random events, $0 < P(A) < 1$, $0 < P(B) < 1$, $P(A|B) + P(\bar{A}|\bar{B}) = 1$, then the correct conclusion is ( ).

   A. A and B are disjoint B. A and B are independent
   C. A and B are NOT independent D. A and B are complementary events

8. Suppose that the pdf of r.v.X is $f(x) = sin(x)$, then the possible value range of X is ( ).

   A. $\left[0, \frac{\pi}{6}\right]$ B. $\left[0, \frac{\pi}{4}\right]$
   C. $\left[0, \frac{\pi}{2}\right]$ D. $\left[0, \frac{2\pi}{3}\right]$

9. If $f(x)$ and $F(x)$ are the probability density function and cumulative distribution function of random variable X, which of the following option is correct? ( )

   A. $P(X = x) = f(x)$ B. $P(X = x) \leq F(x)$
   C. $0 \leq f(x) \leq 1$ D. $P(X = x) = F(x)$

10. Suppose random variable $X \sim N(\mu, \sigma^2)$, $P(X < \mu + \sigma)$ is ( ).

   A. $\Phi(1)$ B. $\Phi(\mu)$
   C. $\Phi(\mu + \sigma)$ D. $\frac{1}{2}$

III. Calculations (6 items, 12 points each)

Score	

11. There are 4 red balls and 6 black balls in a non-perspective bag.

   Calculate:

   (1) If 3 balls were taken randomly one by one without return, what is the probability that 2 black balls and 1 red ball were taken?

   (2) If 3 balls were taken randomly in 3 times, and it will be returned to the bag each time after observed, so what is the probability that 2 black balls were taken for the first 2 times and a red ball was taken for the third time?

12. For a batch of product, 55% are from factory A, 30% are from factory B and the remaining 15% are from factory C. Of these factories, the defective rates are 5%, 20% and 30%, respectively.

Calculate:

(1) The defective rate of this batch of product.

(2) If a defective product was taken randomly, then what is the probability that this product comes from factory C?

13. Suppose that the pdf of r.v. X is $f(x) = \begin{cases} cxe^{-2x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$, where c is a constant.

Calculate:

(1) The value of c;

(2) The cdf of X;

(3) $P(-\frac{1}{2} \leq X < 1)$;

(4) $P(X = 2)$.

14. Suppose that $r.v.$ X take values in 1, 2 ,3, and 4 equally likely, while $r.v.$ Y take integer values among [X, 4]. **Find:**
    (1) The joint probability distribution of (X, Y);
    (2) Two marginal distribution of (X, Y);
    (3) Are X and Y independent to each other?
    (4) The distribution of Z, where $Z = 3X - Y$.

15. If the joint pdf of 2-D r.v. (X, Y) is

$$f(x,y) = \begin{cases} c(x+y), & 0 \le y \le 1 \text{ and } 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases},$$

Calculate:
(1) The constant c;
(2) The marginal distributions of X and Y.
(3) Are X and Y independent to each other?
(4) $P(X < Y)$.

16. The probability density function of random variable X is

$$f_X(x) = \begin{cases} \dfrac{6(2x^2 + x)}{7}, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

The conditional probability density function of random variable Y when given X is

$$f_{Y|X}(y|x) = \begin{cases} \dfrac{2(x^2 + xy)}{2x^2 + x}, & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Calculate:

(1) The joint probability density function of X and Y;

(2) The marginal probability density function of Y;

(3) The conditional probability density function of r.v. X when given Y.

IV. Comprehensive analysis Problem (8 points)

Score	

17. Suppose that r.v. $X \sim N(\mu, \sigma^2)$, proving that $Y = \frac{X-\mu}{\sigma}$ obeys the standard normal distribution.